

Question:

Jen enters a lottery by picking 4 distinct numbers from $S = \{1, 2, 3, \dots, 9, 10\}$. 4 numbers are randomly chosen from S . She wins a prize if at least two of her numbers were 2 of the randomly chosen numbers, and wins the grand prize if all four of her numbers were the randomly chosen numbers. The probability of her winning the grand prize given that she won a prize is $\frac{m}{n}$ where m and n are relatively prime positive integers. Find $m+n$.

Reasoning + Answer:

<think>

1. **Understand the scenario**:

- There are 10 numbers in the set $(S = \{1, 2, 3, \dots, 9, 10\})$.

- Jen picks 4 distinct numbers from (S) .

- She wins a prize if at least 2 of her numbers are among the randomly chosen 4 numbers .

- She wins the grand prize if all 4 of her numbers are among the 4 randomly chosen numbers .

- We need to find the probability that she wins the grand prize given that she won a prize .

2. **Calculate the probability of winning a prize** :

- The total number of ways to choose 4 numbers out of 10 is given by the combination formula $(\binom{10}{4})$:

$$\binom{10}{4} = \frac{10!}{4!(10-4)!} = 210$$

- She wins the grand prize if all 4 of her numbers are among the 4 chosen numbers :

$$\text{Number of ways to choose 4 numbers out of 4} = \binom{4}{4} = 1$$

- She wins a prize if at least 2 of her numbers are among the 4 chosen numbers . We use the complementary probability to find this :

$$P(\text{at least 2 out of 4}) = 1 - P(\text{0 or 1 out of 4})$$

- Probability of picking 0 out of 4 chosen numbers :

$$\text{Number of ways to choose 0 out of 4} = \binom{4}{0} = 1$$

$$P(\text{0 out of 4}) = \frac{1}{210}$$

- Probability of picking 1 out of 4 chosen numbers (using hypergeometric distribution):

$$\text{Number of ways to choose 1 out of 4} = \binom{4}{1} \cdot \binom{6}{3} = 4 \cdot 20 = 80$$

$$P(\text{1 out of 4}) = \frac{80}{210} = \frac{8}{21}$$

- So the probability of at least 2 out of 4 is :

$$P(\text{at least 2 out of 4}) = 1 - \left(\frac{1}{210} + \frac{8}{210}\right) = 1 - \frac{9}{210} = 1 - \frac{3}{70} = \frac{67}{70}$$

3. **Calculate the probability of winning the grand prize given that she won a prize** :

- She wins the grand prize if she picks exactly 4 numbers , and there are $(\binom{4}{4} = 1)$ ways to do this .

- Out of these , the total number of ways to have at least 2 of her numbers among the chosen 4 is $(\binom{10}{4} - 1)$.

- Hence , the probability :

$$P(\text{win grand prize} \mid \text{win prize}) = \frac{\binom{4}{4}}{\binom{10}{4} - 1} = \frac{1}{210 - 1} = \frac{1}{209}$$

4. **Express $(\frac{1}{209})$ as $(\frac{m}{n})$ where (m) and**